

Rally

Product Requirements Document

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1. Overview

RallyDeals is a group-buying e-commerce platform that enables sellers to offer products through two purchasing channels: a group-deal channel, where a discounted price unlocks only once a minimum number of buyers commit within a defined time window, and a normal-listing channel, where products are available at a fixed base price for immediate purchase.

2. Problem Statement / Background

The platform's core differentiator is a risk-free group-deal mechanic: buyers who join a deal are not charged at the time of joining. Instead, a hold is placed on their payment method, and they are only charged if the deal successfully reaches its minimum participation threshold before its time window expires. If the deal fails to reach that threshold, every participant's hold is released, and no one is charged.

This model addresses two connected business needs:

- It gives sellers a low-risk way to validate demand for bulk or promotional pricing before committing to it, supported by referral-driven participant growth to extend reach.
- It gives buyers transparent, risk-free access to group-negotiated discounts, while preserving the option to purchase immediately at base price through the normal-listing channel for buyers who are unwilling or unable to wait for a deal to resolve.

Core Mechanic Summary

A seller lists a product with a single discounted group price, a deal-level stock cap, and a minimum participant count required to unlock the discount. Buyers join the deal's shared pool; the deal succeeds — held payments are captured and orders confirmed — once either the stock cap fills or the timer expires with the minimum met. If the timer expires without the minimum met, the deal fails and every hold is released. Buyers can invite friends via referral links to help fill the pool faster. Each buyer receives their own individual unit — nothing is split or shared. The platform also supports normal, non-group purchases at base price, handled by the same Order/Payment services through a separate, simpler checkout path.

3. Objectives

- Enable sellers to validate product demand before committing to bulk or promotional pricing.
- Enable buyers to access group-negotiated discount prices without upfront financial risk.
- Ensure buyers are never charged for a deal that does not succeed.
- Automate deal resolution (success/failure) with no manual intervention by sellers or administrators.
- Enable referral-driven growth of deals through shareable referral links.
- Provide buyers a no-wait purchasing alternative alongside the group-deal mechanic.
- Provide administrators with visibility into deal outcomes and control over listing quality through an approval workflow.
- Maintain accurate, near real-time visibility into deal progress for buyers.
- Guarantee correctness of stock-cap enforcement under concurrent join attempts.

4. Stakeholders & Roles

A single account may hold both Buyer and Seller roles simultaneously; Admin is a distinct, platform-assigned role.

Stakeholder	Role	Responsibilities
Buyer	End customer	Browses products, joins/leaves group deals, checks out normal purchases, generates/shares referral links, views order history
Seller	End customer (merchant)	Creates and manages product listings, creates and cancels deals, views outcomes of past deals
Admin	Platform operator	Views all deals and their status, approves/rejects product listings, changes user roles
Platform (system actor)	Automated system behavior	Executes deal resolution on timer expiry or stock fill, orchestrates payment/order workflows, dispatches notifications

5. Scope

5.1 In Scope

Per confirmed scope decision, the MVP is the full functional scope defined in the SRS: all 6 formally-scoped epics (Section 6 of this document) and all 51 functional requirements (Section 8).

- Account & Access Management
- Product Catalog Management
- Deal Management (Seller-Facing)
- Group Participation (Buyer-Facing)
- Deal Resolution (Automated)
- Payment Processing
- Normal Purchase (Cart Checkout)
- Order Management
- Notifications
- Admin Moderation & Analytics

5.2 Out of Scope

Explicitly deferred or excluded per the architecture decisions log:

- Multi-tier deal pricing — a seller wanting multiple discount tiers must create multiple separate deals; a single deal with a tier ladder is not supported.
- Service Registry / Discovery (e.g. Eureka) — included in the architecture diagram as a forward-looking piece, but not committed work; built only if time remains after the MVP is functionally complete.
- Kubernetes manifests — same status as above: forward-looking, not committed MVP work.

6. Epics

Each epic below is reproduced from SRS Section 4, including its stated business value and epic-level acceptance criteria.

Account & Access Management

Covers user registration, authentication, and role management, allowing individuals to access the platform as buyers and/or sellers, and allowing admins to manage role assignments.

Business value: Establishes the identity and access foundation required for all buyer, seller, and admin activity on the platform.

Actors: Buyer, Seller, Admin

Epic-level acceptance criteria:

- Users can register and log in using JWT-based authentication.
- A single account can hold both buyer and seller roles.
- Admins can change a user's role.

Product Catalog Management

Covers how sellers create and manage product listings, how newly created listings are gated behind admin approval, and how buyers discover approved products through browsing and search.

Business value: Provides the product inventory foundation on which both normal purchases and group deals are built, while ensuring only compliant listings reach buyers.

Actors: Seller, Buyer, Admin

Epic-level acceptance criteria:

- Sellers can create product listings with name, description, category, and base price; new listings enter a pending-approval state.
- Sellers can update products they own at any time; a product tied to an active deal cannot be deleted but can still be updated.
- Buyers can browse and search (by name, seller, category, price range) only admin-approved products.

Deal Management (Seller-Facing)

Covers a seller's ability to create a group deal on a product they own, cancel a deal before anyone has joined, and review outcomes of past deals.

Business value: Lets sellers validate demand and access group-buy pricing dynamics without committing to bulk discounts until sufficient buyer interest is confirmed.

Actors: Seller

Epic-level acceptance criteria:

- A seller can create a deal on an owned product with a discount price, stock cap, minimum participant count, and duration.
- The minimum participant count can never exceed the stock cap.
- A seller can cancel a deal only if no participant has joined yet.
- A seller can view outcomes of their past deals.

Group Participation (Buyer-Facing)

Covers a buyer's ability to join and leave open deals, track live deal progress, and grow a deal's participation through referral links.

Business value: Delivers the core buyer-facing value proposition of GroupDeal: access to discount pricing through group participation, with transparency and no upfront financial risk.

Actors: Buyer

Epic-level acceptance criteria:

- A buyer can join an open deal if capacity allows and the deal has not resolved, provided they do not already hold an active participation in that deal.
- A buyer can view live deal progress (participants joined vs. target, time remaining).
- A buyer can generate and share a referral link for a deal they joined.
- A buyer can leave a deal only while it is open and more than 10 minutes remain before timer expiry.

Deal Resolution (Automated)

Covers the platform's automatic, system-triggered resolution of deals to success or failure, and the resulting order and payment side effects, with no manual intervention.

Business value: Guarantees buyers and sellers a reliable, tamper-free, and timely outcome for every deal, and ensures the "never charge for a failed deal" guarantee is enforced automatically.

Actors: Platform (system actor)

Epic-level acceptance criteria:

- A deal succeeds when its stock cap fills, or when its timer expires with at least the minimum required participants joined.
- A deal fails when its timer expires with fewer than the minimum required participants joined.
- On success, all pending orders are confirmed, all held payments are captured, and all participants are notified.
- On failure, all pending orders are cancelled, all held payments are voided, reserved inventory is released, and all participants are notified.

Payment Processing

Covers authorization (holds), capture, and voiding of buyer payments for deal purchases, and immediate authorize-and-capture for normal purchases, via Stripe in test mode.

Business value: Ensures buyers are financially protected — never charged unless a deal actually succeeds — while giving sellers and the platform a reliable settlement mechanism.

Actors: Buyer, Platform (system actor)

Epic-level acceptance criteria:

- A pending order is created before payment authorization is requested; a payment method is then authorized (held, not charged) for the discount price.
- A held payment is captured only if the deal succeeds.
- A held payment is voided if the deal fails or the participant leaves.
- Normal purchases are authorized and captured in the same step, with no hold period.
- A declined authorization at join time cancels the pending order and releases the reserved slot; no void occurs, since no hold was ever successfully placed.

Normal Purchase (Cart Checkout)

Covers the non-deal purchase path, where buyers add products to a cart and check out immediately at base price without any waiting period.

Business value: Gives buyers an immediate-purchase alternative for those unwilling or unable to wait for a group deal to resolve.

Actors: Buyer

Epic-level acceptance criteria:

- A buyer can add multiple products to a cart.
- Checkout creates a single order with one line item per product.
- Payment is authorized and captured immediately, and the order is confirmed immediately.

Order Management

Covers a buyer's ability to view their order history and the status of individual orders, covering both deal orders and normal purchases.

Business value: Gives buyers transparency into their purchase history and current order status across both purchase paths.

Actors: Buyer

Epic-level acceptance criteria:

- A buyer can view a history of all their orders (deal and normal).
- A buyer can view the status of each order (pending, confirmed, cancelled).

Notifications

Covers system-generated email notifications sent to buyers at key points: join confirmation, deal outcome, and order confirmation.

Business value: Keeps buyers informed in near real time of the status of their commitments and purchases, reinforcing trust in the platform's automated processes.

Actors: Platform (system actor), Buyer

Epic-level acceptance criteria:

- A buyer receives a join confirmation (email) distinct from a deal-success notification, immediately upon joining a deal.
- A buyer receives a deal outcome notification (email) using a success- or failure-specific template when a deal resolves.
- A buyer receives an order confirmation (email), using a distinct plain-order template, for normal purchases.

Admin Moderation & Analytics

Covers platform administrator capabilities to view all deals and their statuses, and to approve or reject product listings at creation time so that only compliant listings become visible to buyers.

Business value: Gives the platform operational visibility into deal outcomes and a gate that enforces listing quality/compliance before a product reaches buyers.

Actors: Admin

Epic-level acceptance criteria:

- An admin can view all deals and their current status.
- Every newly created product listing is queued for admin approval before it is visible to buyers.
- An admin can change a user's role.

7. User Stories & Acceptance Criteria

Account & Access Management

ID	US-001
Title	User Registration
Epic	Account & Access Management
Story	As a prospective user, I want to register an account, so that I can access the platform as a buyer and/or seller.
Acceptance Criteria	- Given a prospective user provides Name, Email, Phone Number, Password, and a self-selected Role (Buyer or Seller), when they submit registration, then a new account is created with the selected role(s). - Given a prospective user omits a required field or provides invalid data, when they submit registration, then the registration is rejected.
Priority	High
Dependencies	None
Notes	Registration fields are Name, Email, Phone Number, Password, and Role; Role is self-selected. Field-level validation rules (password complexity, phone format, duplicate-email handling) are not specified.

ID	US-002
Title	User Login
Epic	Account & Access Management
Story	As a registered user, I want to log in, so that I can access my account and platform features.
Acceptance Criteria	- Given a registered user provides valid credentials, when they submit a login request, then the system authenticates them and issues a JWT. - Given a registered user provides invalid credentials, when they submit a login request, then authentication is rejected.
Priority	High
Dependencies	US-001
Notes	Authentication is JWT-based.

ID	US-003
Title	Hold Buyer and/or Seller Roles on a Single Account
Epic	Account & Access Management
Story	As a registered user, I want to hold both buyer and seller roles on my account, so that I can both purchase products and list products for sale without maintaining separate accounts.
Acceptance Criteria	Given a registered user account, when the user acts as a buyer or as a seller, then the same account supports both roles concurrently.
Priority	Medium
Dependencies	US-001
Notes	Role is self-selected at registration (US-001). The mechanism by which an existing single-role account later acquires a second role is not specified.

ID	US-004
Title	Admin Changes User Role
Epic	Account & Access Management
Story	As an admin, I want to change a user's role, so that I can manage platform access appropriately.
Acceptance Criteria	Given an admin is viewing a user account, when the admin assigns a new role, then the user's role is updated accordingly.
Priority	Medium
Dependencies	US-001

Product Catalog Management

ID	US-005
Title	Seller Creates a Product Listing
Epic	Product Catalog Management
Story	As a seller, I want to create a product listing, so that buyers can discover and purchase my product.
Acceptance Criteria	Given a seller provides a product name, description, category, and base price, when the seller submits the listing, then a new product listing is created, associated with that seller, and placed in a pending-approval state.
Priority	High
Dependencies	US-001, US-002
Notes	Product creation triggers an admin approval check (US-035); the product is not visible to buyers (US-007) until approved.

ID	US-006
Title	Seller Manages Product Listings
Epic	Product Catalog Management
Story	As a seller, I want to manage my product listings, so that I can keep my catalog accurate and up to date.
Acceptance Criteria	- Given a seller owns a product listing, when the seller updates listing details, then the changes are reflected in the catalog, regardless of whether the product is tied to an active deal. - Given a seller owns a product listing tied to an active deal, when the seller attempts to delete it, then the deletion is rejected. - Given a seller owns a product listing not tied to any active deal, when the seller deletes it, then the listing is removed from the catalog.
Priority	Medium
Dependencies	US-005
Notes	A product tied to an active deal can be updated but not deleted.

ID	US-007
Title	Buyer Browses Products
Epic	Product Catalog Management
Story	As a buyer, I want to browse available products, so that I can discover items to purchase.
Acceptance Criteria	- Given approved products exist in the catalog, when a buyer views the catalog, then approved products are displayed. - Given a product is pending approval or has been rejected, when a buyer views the catalog, then that product is not displayed.
Priority	High
Dependencies	US-005, US-035
Notes	Only admin-approved products are visible to buyers (see US-035).

ID	US-008
Title	Buyer Searches Products
Epic	Product Catalog Management
Story	As a buyer, I want to search for products, so that I can quickly find items I'm interested in.
Acceptance Criteria	Given approved products exist in the catalog, when a buyer submits a search query by name, seller, category, and/or price range, then matching approved products are returned.
Priority	Medium
Dependencies	US-005, US-035
Notes	Search fields are name, seller, category, and price range.

Deal Management (Seller-Facing)

ID	US-009
Title	Seller Creates a Deal
Epic	Deal Management (Seller-Facing)
Story	As a seller, I want to create a group deal on a product I own, so that I can validate demand and offer a discounted price if enough buyers commit.
Acceptance Criteria	- Given a seller owns a product and specifies a discount price, stock cap, minimum participant count, and duration, when the seller submits the deal and the minimum participant count is $\leq$ the stock cap, then the deal is created, the needed stock is reserved from the product's overall inventory, and the deal becomes visible and joinable but not yet running (no timer). - Given the specified minimum participant count exceeds the stock cap, when the seller submits the deal, then the deal creation is rejected.
Priority	High
Dependencies	US-005

ID	US-010
Title	Seller Cancels a Deal
Epic	Deal Management (Seller-Facing)
Story	As a seller, I want to cancel a deal I created, so that I can withdraw an offer before any buyer has committed to it.
Acceptance Criteria	- Given a deal has zero participants joined, when the seller cancels the deal, then the deal is cancelled. - Given a deal has one or more participants joined, when the seller attempts to cancel the deal, then the cancellation is rejected.
Priority	Medium
Dependencies	US-009

ID	US-011
Title	Seller Views Deal Outcomes
Epic	Deal Management (Seller-Facing)
Story	As a seller, I want to view the outcomes of my past deals, so that I can understand how my offers performed.
Acceptance Criteria	Given a seller has one or more resolved deals, when the seller views deal history, then the outcome (success/failure) of each past deal is displayed.
Priority	Low
Dependencies	US-009

Group Participation (Buyer-Facing)

ID	US-012
Title	Buyer Joins an Open Deal
Epic	Group Participation (Buyer-Facing)
Story	As a buyer, I want to join an open deal, so that I can access the discounted group price without being charged unless the deal succeeds.
Acceptance Criteria	- Given a deal has open capacity and has not resolved, when a buyer requests to join, then the system checks for a free slot; if available, the buyer is listed as a participant, a slot is reserved, a pending order is created, — if this is the first participant — the deal's timer starts, and a payment authorization hold is triggered (US-022) for the discount price, requested only after the participant is listed and the pending order exists. - Given a deal is full or already resolved, when a buyer requests to join, then the join is rejected with no side effects. - Given the payment authorization triggered by this join is declined, when the decline is reported back, then the order-cancellation-and-slot-release behavior in US-025 is invoked. - Given a buyer already has an active participation in a deal, when that buyer attempts to join the same deal again, then the second join is rejected.
Priority	High
Dependencies	US-009, US-022, US-025

ID	US-013
Title	Buyer Views Live Deal Progress
Epic	Group Participation (Buyer-Facing)
Story	As a buyer, I want to view live progress of a deal, so that I can decide whether to join and track how close it is to succeeding.
Acceptance Criteria	Given an active deal, when a buyer views the deal, then current participant count vs. target and time remaining are displayed, updated in near real time.
Priority	High
Dependencies	US-009

ID	US-014
Title	Buyer Generates and Shares a Referral Link
Epic	Group Participation (Buyer-Facing)
Story	As a participant in a deal, I want to generate and share a referral link, so that I can invite others to join and help the deal succeed.
Acceptance Criteria	Given a buyer has joined a deal, when the buyer requests a referral link, then a shareable link tied to that specific deal is generated.
Priority	Medium
Dependencies	US-012

ID	US-015
Title	New Buyer Joins via Referral Link
Epic	Group Participation (Buyer-Facing)
Story	As a buyer who received a referral link, I want to join the deal through that link, so that I can participate, with the referrer credited for the referral.
Acceptance Criteria	Given a valid referral link for an open deal, when a buyer opens the link and joins, then the buyer goes through the standard join flow, and the system records who referred them for tracking purposes.
Priority	Medium
Dependencies	US-012, US-014

ID	US-016
Title	Buyer Leaves a Deal
Epic	Group Participation (Buyer-Facing)
Story	As a participant in a deal, I want to leave a deal I previously joined, so that I can withdraw my commitment if I change my mind, while the deal is still open.
Acceptance Criteria	- Given a deal is still open and more than 10 minutes remain before timer expiry, when the participant requests to leave, then their slot is released, their order is cancelled, and the void-hold behavior in US-024 is triggered. - Given the deal has already resolved, or fewer than 10 minutes remain before timer expiry, when the participant requests to leave, then the request is rejected.
Priority	High
Dependencies	US-012, US-024
Notes	Ownership boundary: this story owns eligibility checking, slot release, and order cancellation; the actual payment-void logic is owned by US-024.

Deal Resolution (Automated)

ID	US-017
Title	Automatic Deal Success on Stock Cap Fill
Epic	Deal Resolution (Automated)
Story	As the platform, I want to automatically resolve a deal as successful the instant its stock cap fills, so that buyers and sellers get a timely, reliable outcome without manual intervention.
Acceptance Criteria	Given a deal's stock cap becomes fully filled, when the last slot is taken, then the deal resolves as successful immediately, even if time remains.
Priority	High
Dependencies	US-012

ID	US-018
Title	Automatic Deal Success on Timer Expiry with Minimum Met
Epic	Deal Resolution (Automated)
Story	As the platform, I want to automatically resolve a deal as successful when its timer expires with at least the minimum required participants, so that the deal succeeds as intended even if the stock cap was never fully filled.
Acceptance Criteria	Given a deal's timer expires, when the number of joined participants is at least the minimum required, then the deal resolves as successful.
Priority	High
Dependencies	US-012

ID	US-019
Title	Automatic Deal Failure on Timer Expiry Below Minimum
Epic	Deal Resolution (Automated)
Story	As the platform, I want to automatically resolve a deal as failed when its timer expires with fewer than the minimum required participants, so that buyers are protected from being charged for an under-subscribed deal.
Acceptance Criteria	Given a deal's timer expires, when the number of joined participants is fewer than the minimum required, then the deal resolves as failed.
Priority	High
Dependencies	US-012

ID	US-020
Title	System Processes Deal-Success Side Effects
Epic	Deal Resolution (Automated)
Story	As the platform, I want to confirm orders when a deal succeeds and trigger payment capture and participant notification, so that buyers and sellers receive the correct outcome consistently.
Acceptance Criteria	- Given a deal resolves as successful, when resolution is processed, then every pending order tied to the deal is confirmed. - Given orders have been confirmed for a successful deal, when the success workflow completes, then the capture behavior in US-023 and the notification behavior in US-032 are triggered.
Priority	High
Dependencies	US-017, US-018, US-023, US-032
Notes	Ownership boundary: this story owns order confirmation only; capture is owned by US-023 and notification by US-032.

ID	US-021
Title	System Processes Deal-Failure Side Effects
Epic	Deal Resolution (Automated)
Story	As the platform, I want to cancel orders and release inventory when a deal fails, and trigger payment voiding and participant notification, so that no buyer is charged and reserved inventory becomes available again.
Acceptance Criteria	- Given a deal resolves as failed, when resolution is processed, then every pending order tied to the deal is cancelled and reserved inventory is released back to general stock. - Given orders have been cancelled and inventory released for a failed deal, when the failure workflow completes, then the void behavior in US-024 and the notification behavior in US-032 are triggered.
Priority	High
Dependencies	US-019, US-024, US-032
Notes	Ownership boundary: this story owns order cancellation and inventory release only; voiding is owned by US-024 and notification by US-032.

Payment Processing

ID	US-022
Title	Authorize Payment Hold at Join
Epic	Payment Processing
Story	As the platform, I want to place a payment authorization hold when a buyer joins a deal, so that funds are reserved without charging the buyer upfront.
Acceptance Criteria	Given a buyer successfully joins a deal, when the join is processed, then a payment hold (authorization) for the discount price is placed via Stripe test mode, without a charge occurring.
Priority	High
Dependencies	US-012
Notes	Uses Stripe in test mode (test API keys, test card numbers); no real money moves.

ID	US-023
Title	Capture Payment on Deal Success
Epic	Payment Processing
Story	As the platform, I want to capture a held payment when a deal succeeds, so that the buyer is charged only after the deal is confirmed to succeed.
Acceptance Criteria	Given a deal resolves as successful, when resolution is processed, then each associated payment hold is captured, actually charging the buyer.
Priority	High
Dependencies	US-020

ID	US-024
Title	Void Payment Hold
Epic	Payment Processing
Story	As the platform, I want to void a held payment when a deal fails or a participant leaves, so that a buyer is never charged for a commitment that did not result in a purchase.
Acceptance Criteria	Given a deal fails (triggered by US-021) or a participant leaves a deal (triggered by US-016), when the relevant event occurs, then the associated payment hold is voided and no charge is made.
Priority	High
Dependencies	US-016, US-021
Notes	A declined join-time authorization is handled by US-025 (order cancellation and slot release), not by this story, since no hold is ever successfully placed before a decline.

ID	US-025
Title	Declined Authorization Cancels Order and Releases Reserved Slot
Epic	Payment Processing
Story	As the platform, I want to cancel the pending order and release the reserved slot when a payment authorization is declined at join time, so that the slot doesn't become permanently unavailable and no orphaned pending order remains.
Acceptance Criteria	Given a buyer's payment authorization is declined during a join attempt, when the decline is detected, then the pending order created for that join is set to cancelled, and the reserved slot is released back to the pool for other buyers.
Priority	High
Dependencies	US-012
Notes	Because the pending order is created before authorization is requested (US-012), a decline cancels that order in addition to freeing the slot. No payment void occurs here.

ID	US-026
Title	Immediate Authorize and Capture for Normal Purchase
Epic	Payment Processing
Story	As a buyer, I want my payment to be authorized and captured immediately at checkout for a normal purchase, so that I don't have to wait for a deal to resolve.
Acceptance Criteria	Given a buyer checks out a cart of normal (non-deal) products, when checkout is processed, then payment is authorized and captured in the same step, and the order is immediately confirmed.
Priority	High
Dependencies	US-028

Normal Purchase (Cart Checkout)

ID	US-027
Title	Buyer Adds Products to Cart
Epic	Normal Purchase (Cart Checkout)
Story	As a buyer, I want to add multiple products to a cart, so that I can purchase several items in a single transaction.
Acceptance Criteria	Given products available at base price, when a buyer adds one or more products to the cart, then the cart reflects the selected products.
Priority	High
Dependencies	US-007

ID	US-028
Title	Buyer Checks Out Cart
Epic	Normal Purchase (Cart Checkout)
Story	As a buyer, I want to check out my cart as a single order, so that I can complete a normal purchase without waiting.
Acceptance Criteria	- Given a buyer's cart contains one or more products, when the buyer checks out, then a single order is created with one line item per product, and the immediate authorize-and-capture behavior in US-026 is triggered. - Given the triggered payment succeeds, when authorization and capture complete, then the order is confirmed.
Priority	High
Dependencies	US-027, US-026
Notes	Ownership boundary: this story owns cart-to-order translation and order confirmation; authorize-and-capture logic is owned by US-026.

Order Management

ID	US-029
Title	Buyer Views Order History
Epic	Order Management
Story	As a buyer, I want to view my order history, so that I can review my past deal orders and normal purchases.
Acceptance Criteria	Given a buyer has one or more orders (deal or normal), when the buyer views order history, then all their orders are listed.
Priority	Medium
Dependencies	US-012, US-028

ID	US-030
Title	Buyer Views Order Details
Epic	Order Management
Story	As a buyer, I want to view the details of an order, so that I can review its information and current status.
Acceptance Criteria	- Given a buyer selects an order, when the order details page opens, then the order status is displayed. - Then purchased items are displayed with quantities and prices. - Then order date and total amount are displayed.
Priority	Medium
Dependencies	US-029

Notifications

ID	US-031
Title	Join Confirmation Notification
Epic	Notifications
Story	As a buyer, I want to receive a join confirmation when I join a deal, so that I know my join was recorded, without assuming the deal has succeeded.
Acceptance Criteria	Given a buyer successfully joins a deal, when the join is processed, then the buyer receives an email join confirmation distinct from a deal-success notification.
Priority	Medium
Dependencies	US-012

ID	US-032
Title	Deal Outcome Notification
Epic	Notifications
Story	As a buyer, I want to receive a notification when a deal I joined resolves, so that I know whether it succeeded or failed.
Acceptance Criteria	- Given a deal resolves as successful, when resolution is processed, then all participants receive an email outcome notification, near real time, using a success-specific template that includes order information plus a success acknowledgement. - Given a deal resolves as failed, when resolution is processed, then all participants receive an email outcome notification, near real time, using a failure-specific template that includes order information plus an explanation that the deal did not succeed.
Priority	Medium
Dependencies	US-020, US-021
Notes	The deal outcome notification uses a distinct template from the order confirmation (US-033): both include order information, but this one adds success/failure-specific messaging.

ID	US-033
Title	Order Confirmation Notification
Epic	Notifications
Story	As a buyer, I want to receive an order confirmation, so that I know my purchase was successfully completed.
Acceptance Criteria	Given a normal purchase order is confirmed, when checkout completes, then the buyer receives an email order confirmation.
Priority	Medium
Dependencies	US-028
Notes	This is a plain order-information email, distinct from the deal outcome notification (US-032). Deal-order confirmations are covered by US-032's success template rather than a separate confirmation email.

Admin Moderation & Analytics

ID	US-034
Title	Admin Views All Deals
Epic	Admin Moderation & Analytics
Story	As an admin, I want to view all deals and their status, so that I have visibility into platform deal activity.
Acceptance Criteria	Given deals exist on the platform, when an admin views the deals list, then all deals and their current status are displayed.
Priority	Medium
Dependencies	US-009

ID	US-035
Title	Admin Approves or Rejects Product Listings
Epic	Admin Moderation & Analytics
Story	As an admin, I want to approve or reject a product listing when it is created, so that only compliant listings become visible to buyers.
Acceptance Criteria	- Given a seller creates a new product listing (US-005), when the listing is submitted, then it enters a pending-approval state and is queued for admin review. - Given an admin reviews a pending product listing, when the admin approves it, then the listing becomes visible to buyers (US-007, US-008). - Given an admin reviews a pending product listing, when the admin rejects it, then the listing remains hidden from buyers.
Priority	Medium

8. Functional Requirements

Full functional requirements traceability matrix, from SRS Section 6.

FR ID	Requirement	Traceability
FR-001	The system shall allow a prospective user to register a new account by providing Name, Email, Phone Number, Password, and a self-selected Role (Buyer or Seller).	US-001
FR-002	The system shall authenticate registered users using JWT-based login.	US-002
FR-003	The system shall support a single account holding both buyer and seller roles concurrently.	US-003
FR-004	The system shall allow an admin to change a user's role.	US-004
FR-005	The system shall allow a seller to create a product listing with name, description, category, and base price.	US-005
FR-005a	The system shall place a newly created product listing into a pending-approval state, triggering the admin approval workflow.	Owner: US-035; triggered by US-005
FR-006	The system shall allow a seller to update a product listing they own at any time, regardless of whether it is tied to an active deal.	US-006
FR-006a	The system shall reject deletion of a product listing that is tied to an active deal.	US-006
FR-006b	The system shall allow deletion of a product listing that is not tied to any active deal.	US-006
FR-007	The system shall allow a buyer to browse only admin-approved product listings in the catalog.	US-007
FR-008	The system shall allow a buyer to search the catalog of admin-approved products by name, seller, category, and/or price range.	US-008
FR-009	The system shall allow a seller to create a deal on a product they own, specifying discount price, stock cap, minimum participant count, and duration.	US-009
FR-010	The system shall reject deal creation if the minimum participant count exceeds the stock cap.	US-009
FR-011	The system shall reserve the deal's needed stock from the product's overall inventory at deal creation.	US-009
FR-012	The system shall make a newly created deal visible and joinable, without starting its timer, until the first join occurs.	US-009
FR-013	The system shall allow a seller to cancel a deal only if no participant has joined it.	US-010
FR-014	The system shall reject a seller's cancellation request for a deal that has one or more participants.	US-010
FR-015	The system shall allow a seller to view the outcomes of their past deals.	US-011
FR-016	The system shall allow a buyer to join an open, unresolved deal with available capacity.	US-012

FR ID	Requirement	Traceability
FR-017	The system shall check for a free slot, then reserve one slot and list the buyer as a participant, before creating that participant's pending order.	US-012
FR-018	The system shall start a deal's timer at the moment its first participant joins.	US-012
FR-019	The system shall create a pending order for the joining buyer before requesting payment authorization.	US-012
FR-019a	The system shall authorize (hold, not charge) a buyer's payment method for the discount price, triggered after the pending order is created for that join.	Owner: US-022; triggered by US-012
FR-020	The system shall reject a join request with no side effects if the deal is full or already resolved.	US-012
FR-021	The system shall reject a second, concurrent join request from a buyer who already has an active participation in that deal.	US-012
FR-021a	The system shall allow a buyer to join a deal for a product they have already purchased through a normal order, and to rejoin a deal they previously left.	US-012
FR-022	The system shall cancel the pending order and release the reserved slot back to the pool if the payment authorization for that join is declined	Owner: US-025; triggered by US-012
FR-023	The system shall display live deal progress (participants joined vs. target, time remaining) to buyers viewing an active deal, updated in near real time.	US-013
FR-024	The system shall allow a participant in a deal to generate a shareable referral link tied to that specific deal.	US-014
FR-025	The system shall route new participants who join via a referral link through the standard join flow.	US-015
FR-026	The system shall record the referrer for a buyer who joins via a referral link, for tracking purposes.	US-015
FR-027	The system shall allow a participant to leave a deal only while it is open and more than 10 minutes remain before timer expiry.	US-016
FR-028	The system shall reject a leave request if the deal has resolved or fewer than 10 minutes remain before timer expiry.	US-016
FR-029	The system shall, upon a valid leave request, release the participant's slot and cancel their order.	US-016
FR-029a	The system shall void a participant's payment hold, triggered when they leave a deal.	Owner: US-024; triggered by US-016
FR-030	The system shall automatically resolve a deal as successful the instant its stock cap fills, even if time remains.	US-017
FR-031	The system shall automatically resolve a deal as successful when its timer expires and at least the minimum required participants have joined.	US-018
FR-032	The system shall automatically resolve a deal as failed when its timer expires with fewer than the minimum required participants joined.	US-019

FR ID	Requirement	Traceability
FR-033	The system shall, upon deal success, confirm every pending order tied to the deal.	US-020
FR-034	The system shall capture every associated payment hold, triggered upon deal success.	Owner: US-023; triggered by US-020
FR-035	The system shall notify all participants of the outcome, triggered upon deal success.	Owner: US-032; triggered by US-020
FR-036	The system shall, upon deal failure, cancel every pending order tied to the deal.	US-021
FR-037	The system shall void every associated payment hold, triggered upon deal failure.	Owner: US-024; triggered by US-021
FR-038	The system shall, upon deal failure, release the deal's reserved inventory back to general stock.	US-021
FR-039	The system shall notify all participants of the outcome, triggered upon deal failure.	Owner: US-032; triggered by US-021
FR-040	The system shall authorize and capture payment in the same step for normal (non-deal) purchases.	US-026
FR-041	The system shall allow a buyer to add one or more products to a cart.	US-027
FR-042	The system shall allow a buyer to check out a cart as a single order with one line item per product.	US-028
FR-043	The system shall immediately confirm a normal purchase order upon successful checkout.	US-028
FR-044	The system shall allow a buyer to view their order history, covering both deal orders and normal purchases.	US-029
FR-045	The system shall allow a buyer to view the status of an individual order (pending, confirmed, or cancelled).	US-030
FR-046	The system shall send a buyer an email join confirmation upon successfully joining a deal, distinct from a deal-success notification.	US-031
FR-047	The system shall send all participants an email deal outcome notification when a deal resolves, using a success-specific template on success and a failure-specific template on failure, each including order information plus outcome-specific messaging.	US-032
FR-048	The system shall send a buyer an email order confirmation, using a plain order-information template distinct from the deal outcome templates, for normal purchases.	US-033
FR-049	The system shall allow an admin to view all deals and their current status.	US-034
FR-050	The system shall queue every newly created product listing for admin review before it becomes visible to buyers.	US-035
FR-050a	The system shall allow an admin to approve a pending product listing, making it visible to buyers.	US-035

FR ID	Requirement	Traceability
FR-050b	The system shall allow an admin to reject a pending product listing, keeping it hidden from buyers.	US-035
FR-051	The system shall ensure each buyer receives their own individual unit of a product on deal success; group participation aggregates demand only, and nothing is split or shared between participants.	Rules & Constraints

9. Non-Functional Requirements

Performance

ID	Requirement
NFR-001	Live deal progress (participant count, time remaining) shall update in near real time for anyone viewing an active deal.
NFR-002	Buyers shall receive deal outcome notifications (success/failure) close to real time.

Reliability

ID	Requirement
NFR-003	The system shall never charge a buyer for a deal that did not succeed.
NFR-004	Deal outcomes shall resolve automatically, with no manual trigger required from a seller or admin.

Data Integrity / Concurrency

ID	Requirement
NFR-005	The system shall never allow more participants to join a deal than its stock cap, even under concurrent join attempts.

Security

ID	Requirement
NFR-006	The system shall use JWT-based authentication for user access.
NFR-007	Security requirements beyond JWT-based authentication (password policy, encryption standards, session/token expiry, rate limiting) are not currently defined.

Scalability

ID	Requirement
NFR-008	The system shall support at least 50 concurrent users browsing/interacting with the platform with API response times under 2 seconds, as validated via local load testing.

Availability

ID	Requirement
NFR-009	Availability/uptime targets are not currently defined.

Usability

ID	Requirement
NFR-010	Usability requirements (accessibility standards, supported devices/browsers, UI response-time expectations) are not currently defined.

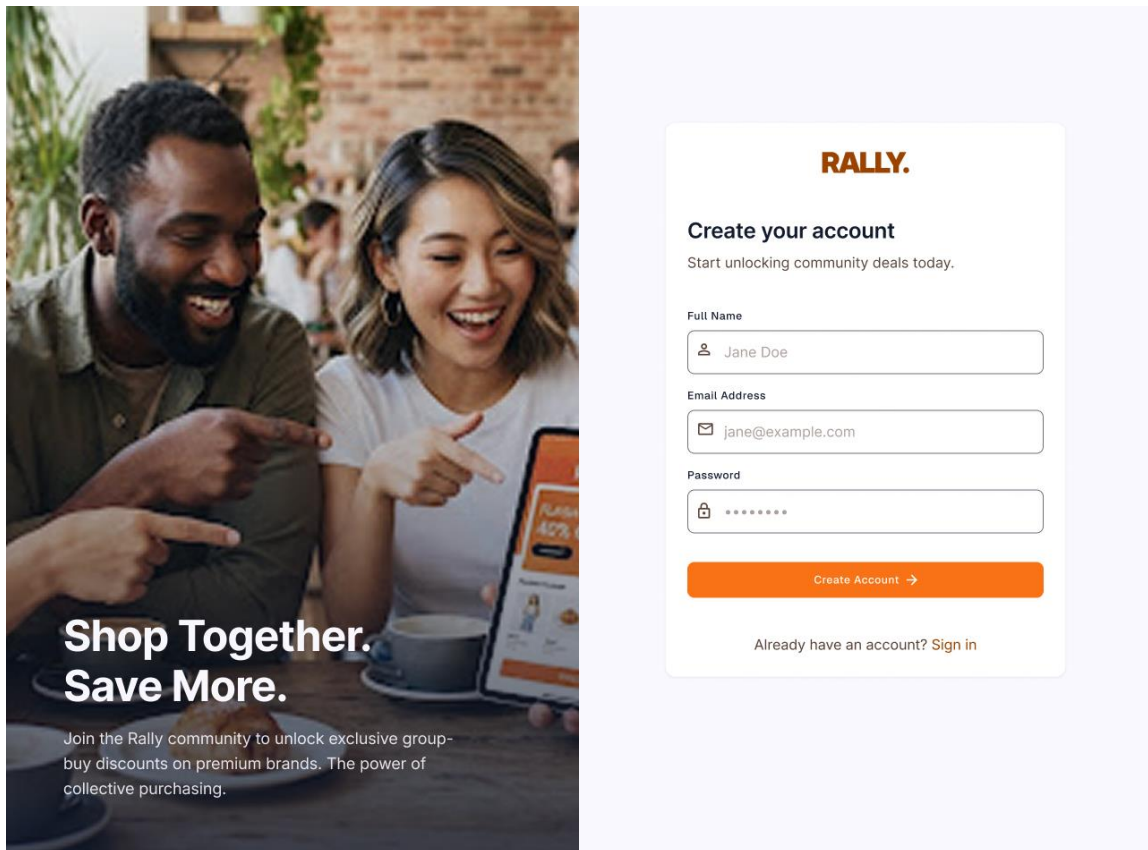
Maintainability

ID	Requirement
NFR-011	The system's payment/order/inventory workflows shall be implemented using sagas — sequences of coordinated steps across services with defined compensating actions if a later step fails — to support maintainable, recoverable distributed transactions.

10. System Services

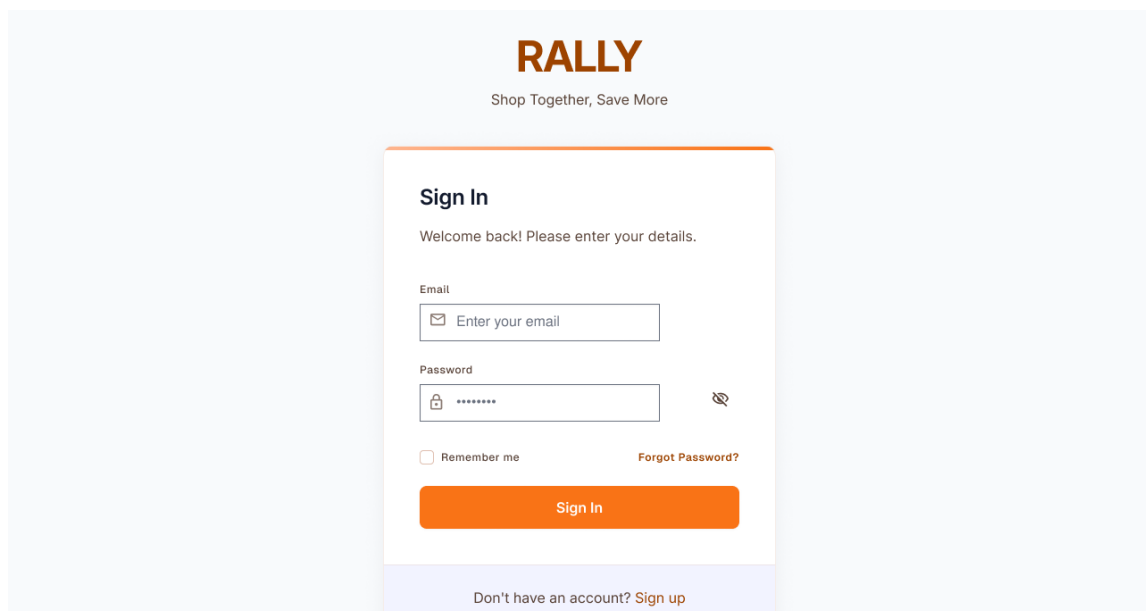
Service	Responsibility	DB
API Gateway	Single entry point, routing, JWT validation	—
Auth Service	Registration, login, JWT issuance, roles	Postgres
Catalog Service	Product listings, categories, search	Postgres
Deal Service	Deal lifecycle, pricing, capacity, timer bookkeeping (merged former Scheduler Service), single source of truth for deal state	Postgres
Participation Service	Join/leave, referral links, high-write participant tracking	Postgres
Inventory Service	Product-level stock reservation for deals	Postgres
Order Service	Owns all orders (deal + normal); the only service whose payment needs Payment Service acts on — relationship is purely event-driven (Kafka), no synchronous calls either direction	Postgres
Payment Service	Stripe (test mode) integration: authorize / capture / void	Postgres
Notification Service	Kafka consumer → WebSocket/email push	

11. Wireframes



The Register Page wireframe is divided into two main sections. The left section features a background image of a man and a woman smiling and pointing at a tablet. Overlaid on this image is the text "Shop Together. Save More." in a large, bold, white font. Below this, in a smaller white font, is the text: "Join the Rally community to unlock exclusive group-buy discounts on premium brands. The power of collective purchasing." The right section is a white card with a light orange border. At the top of the card is the "RALLY." logo in orange. Below the logo is the heading "Create your account" in bold black text, followed by the subtext "Start unlocking community deals today." in a smaller black font. The card contains three input fields: "Full Name" with a person icon and the placeholder "Jane Doe", "Email Address" with an envelope icon and the placeholder "jane@example.com", and "Password" with a lock icon and placeholder dots. Below these fields is an orange button with the text "Create Account" and a right-pointing arrow. At the bottom of the card, there is a link that says "Already have an account? Sign in".

Figure 1. Register Page



The Login Page wireframe has a light blue background. At the top center is the "RALLY" logo in orange, with the tagline "Shop Together, Save More" in a smaller black font below it. In the center of the page is a white card with an orange border. The card has the heading "Sign In" in bold black text, followed by the subtext "Welcome back! Please enter your details." in a smaller black font. Below this are two input fields: "Email" with an envelope icon and placeholder "Enter your email", and "Password" with a lock icon, placeholder dots, and a toggle icon to the right. Below the password field are two links: "Remember me" with an unchecked checkbox, and "Forgot Password?". At the bottom of the card is an orange button with the text "Sign In". Below the card, centered on the light blue background, is a link that says "Don't have an account? Sign up".

Figure 2. Login Page

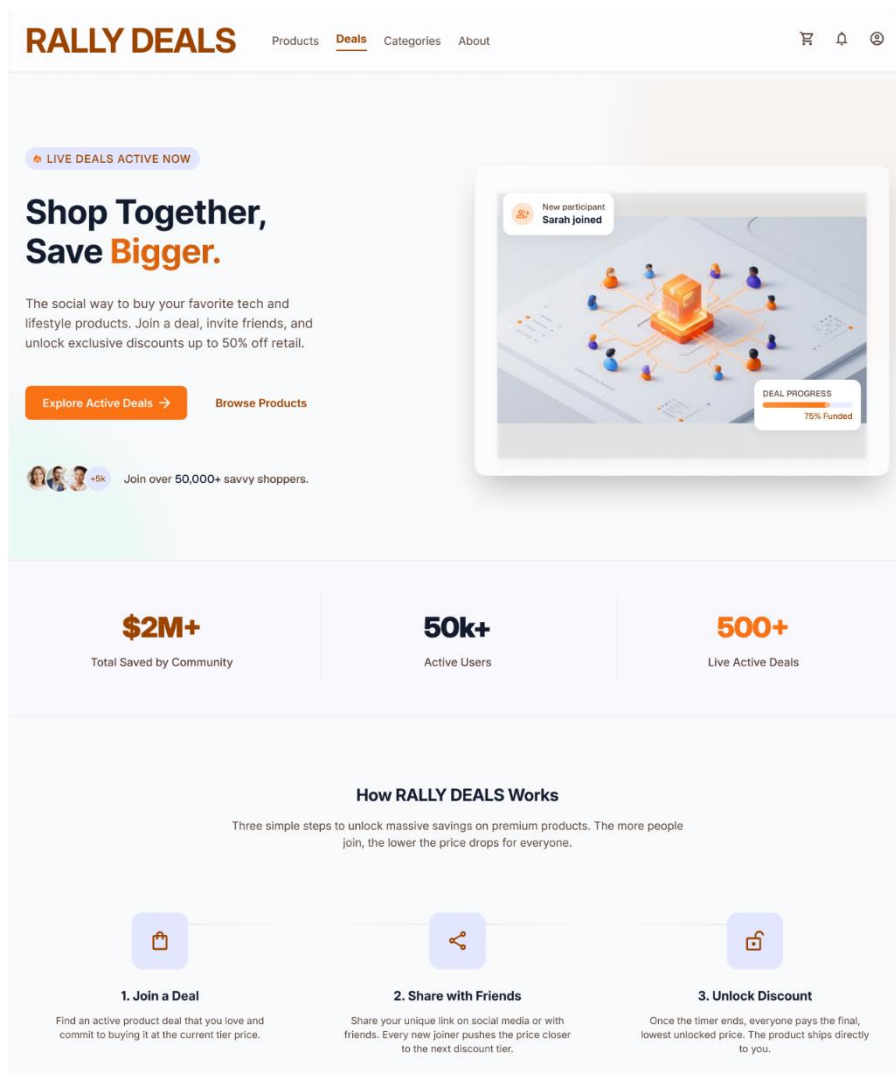


Figure 3. Landing Page View 1

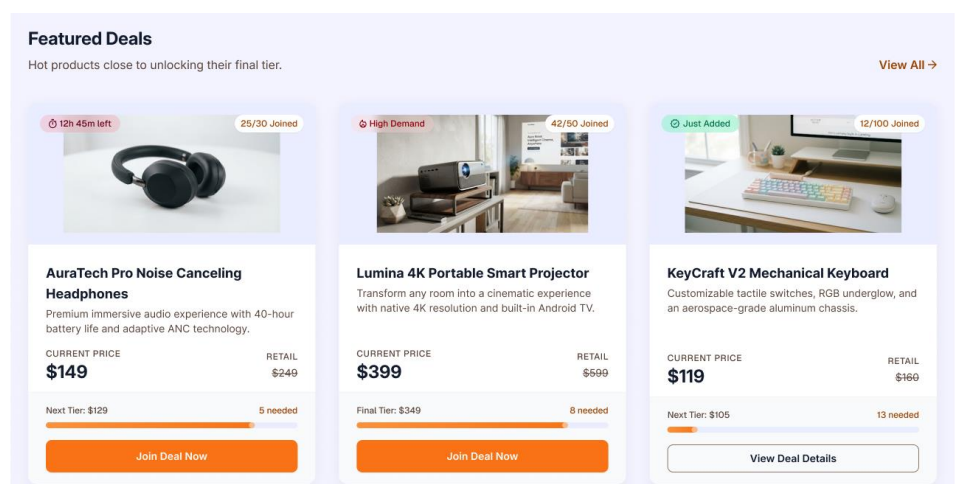


Figure 4. Landing Page View 2

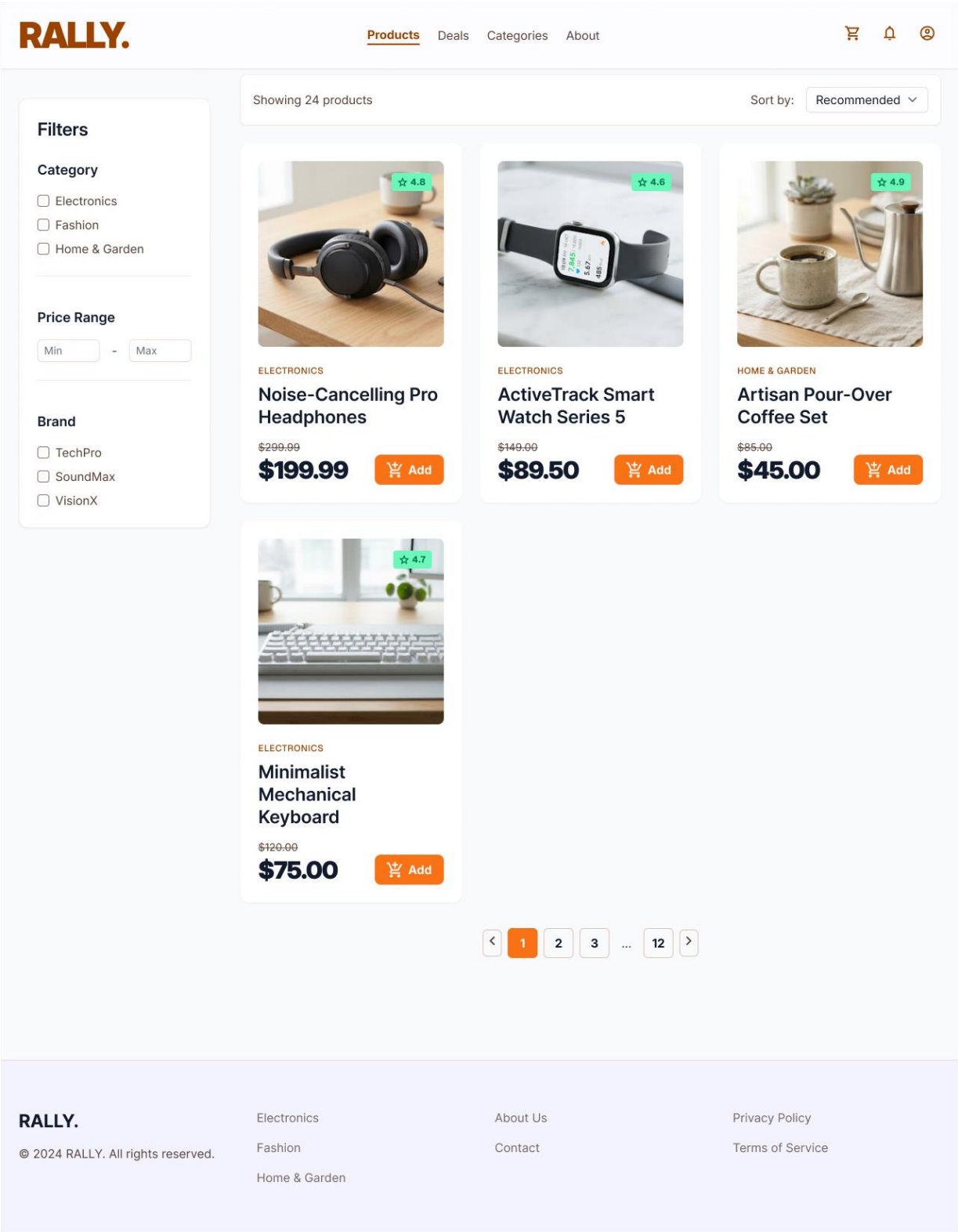


Figure 5. Browsing Products Page

Home > Electronics > Peripherals > Premium Mechanical Keyboard



RALLY ACTIVE. Selling Fast

Nimbus Pro Mechanical Keyboard

Hot-swappable, gasket mounted, full aluminum chassis. The ultimate typing experience.

RALLY PRICE. **\$149** ~~\$249~~ **ENDS IN 04:12:39:55**

142 Joined Target: 150
Only 8 spots left at this price!

+139 just joined

Join This Rally. →

Invite Friends

- Secure Payment
- Free Ship Over \$50
- 30-Day Returns

Description Specifications Live Activity FAQ

The Nimbus Pro represents the pinnacle of tactile feedback and acoustic dampening. Designed for enthusiasts and professionals alike, this keyboard transforms your daily workflow into a sensory experience.

Premium Construction

Machined from a single block of aerospace-grade aluminum, the chassis provides unparalleled rigidity and weight. The gasket-mounted design isolates the plate from the case, resulting in a slightly bouncy typing feel and a refined, muted sound profile devoid of harsh pings.

Hot-Swappable PCB
Easily change switches without soldering to customize your feel.

Acoustic Dampening
Pre-installed silicone dampeners for a deep, satisfying "thock".

Deal Milestones



RALLY.

Shop Together, Save More. Premium deals curated for the community.

Categories

- Electronics
- Fashion
- Home & Garden

Company

- About Us
- Contact

Legal

- Privacy Policy
- Terms of Service

Figure 6. Deal Details Page

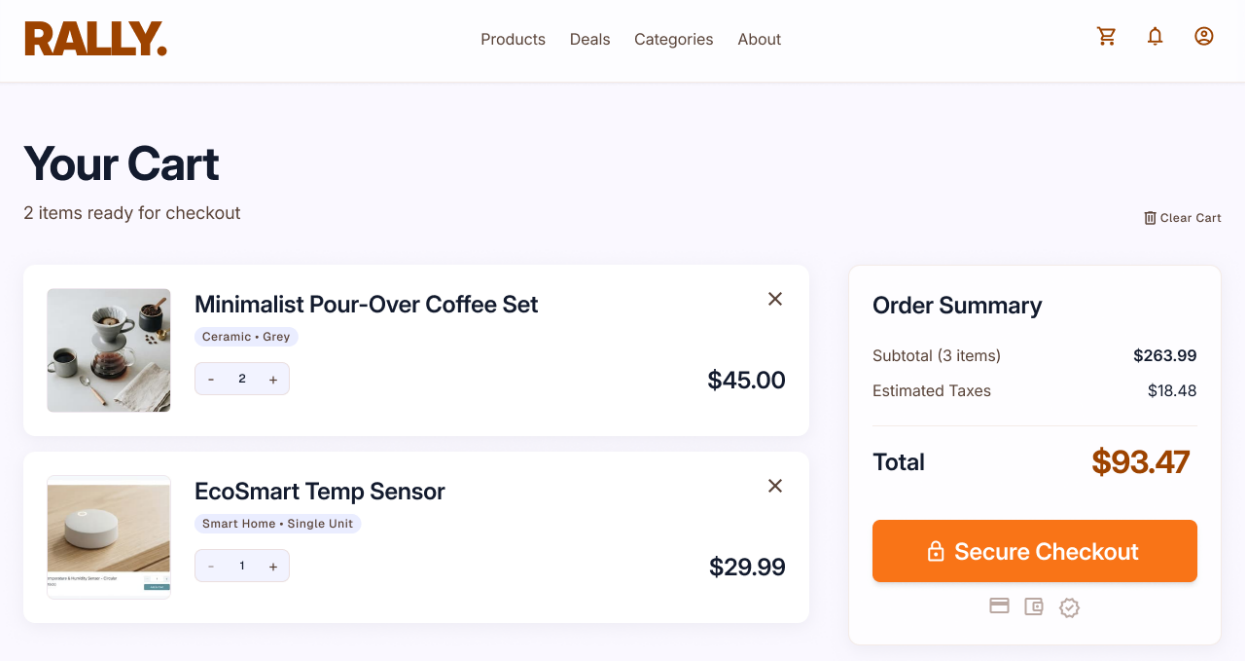


Figure 7. Cart Page

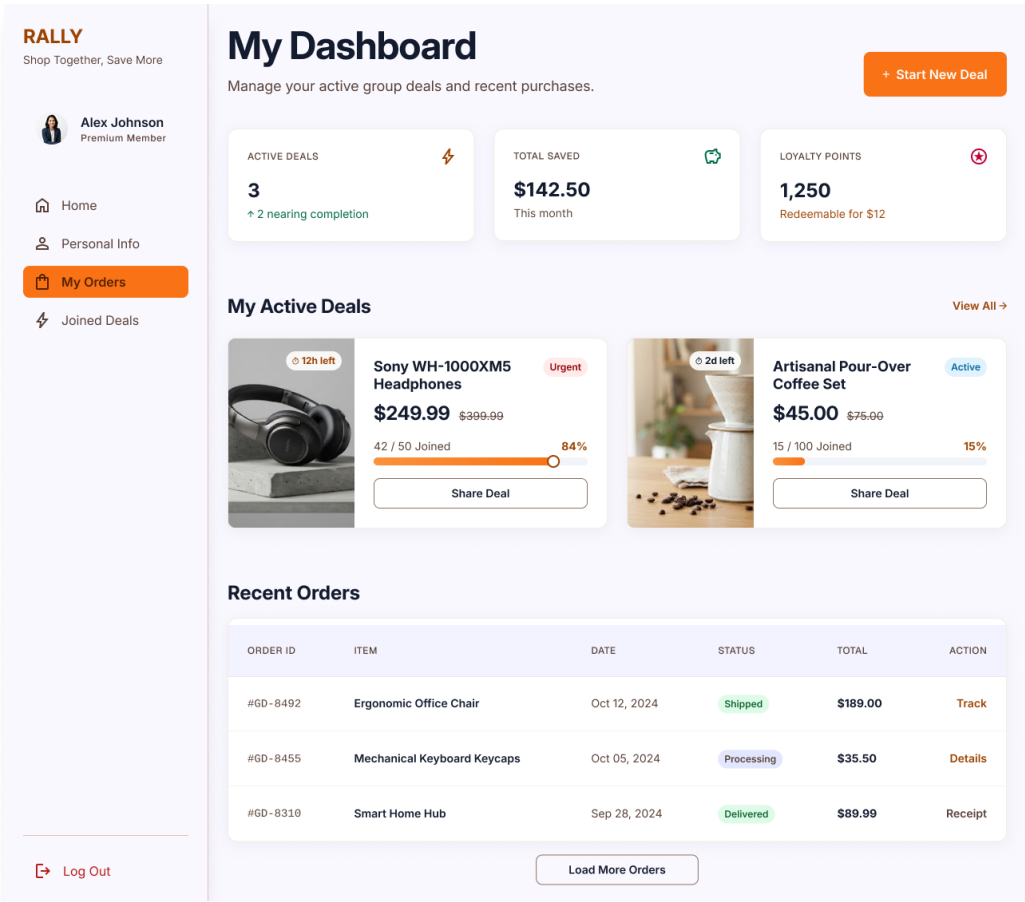


Figure 8. User Profile Page

12. Use Case

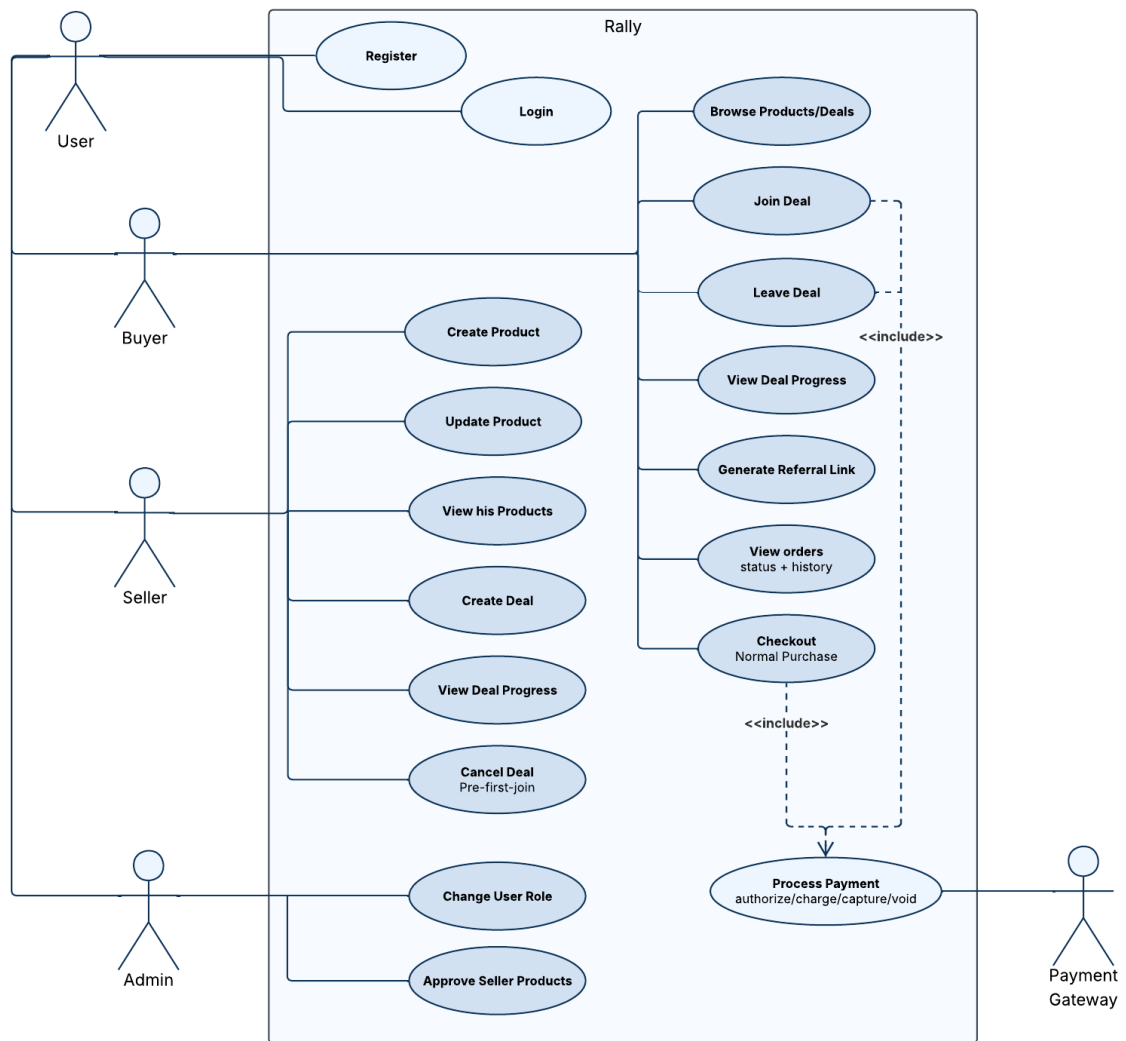


Figure 9. Use Case Diagram

13. Component Diagram

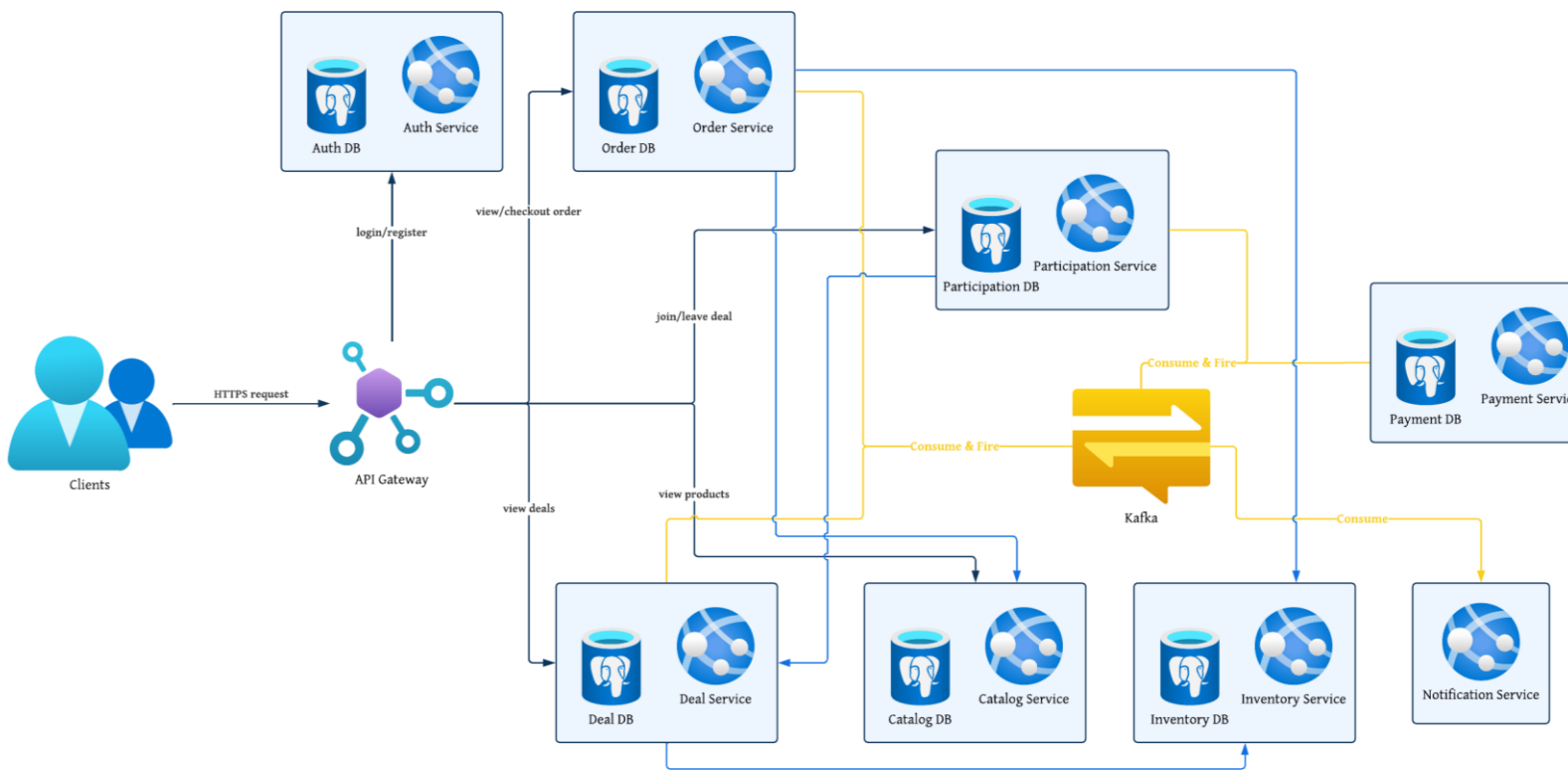


Figure 10. Component Diagram

14. ERD

Figure 11. ERD Diagram